

No significant effect of caffeine on sustained attention in older adults

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ABSTRACT

Whether caffeine benefits attention in later life is unclear. We tested a 200 mg caffeine dose against placebo in community-dwelling older adults. Caffeine did not significantly change sustained attention scores relative to placebo. Routine caffeine supplementation to support attention in older adults is not supported by these data.

METHODS

A double-blind crossover design enrolled n = 85 older adults (ages 65-82, 47 female). Sustained attention was measured with the Psychomotor Vigilance Task (PVT) 45 minutes after each administration, one week apart.

RESULTS

Caffeine did not significantly change sustained attention scores compared with placebo (mean difference 0.6 points, 95% CI -1.1 to 2.3, $p = 0.41$). Self-reported alertness increased modestly ($p = 0.04$) without a corresponding performance change.

DISCUSSION

In contrast with reports in younger samples, caffeine produced no measurable sustained-attention benefit in older adults. Age-related differences in adenosine signalling or habitual intake may explain the discrepancy.

LIMITATIONS

Crossover design with a single dose; habitual caffeine intake was self-reported.